

Econometrics I

Lecture 9a: Difference-in-Differences

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Fall 2026

Recall: Fundamental Problem of Causal Inference

Fundamental Problem of Causal Inference

- ▶ We don't observe the **counterfactual** $Y_i(1 - D_i)$.
- ▶ For a single individual we either observe $Y_i(1)$ **or** $Y_i(0)$ but never both!
 - ex: We don't see what your wage would have been if you didn't attend college.
 - ex: We might know your cholesterol before you took Lipitor, but we don't know what it would be today if you didn't take Lipitor.

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

i	$Y_i(1)$	$Y_i(0)$	D_i	X_i
1	1	?	1	3.2
2	0	?	1	1.7
3	?	0	0	2.5
		$\vdots$		
n	?	1	0	-1.5

How do we construct the counterfactual?

We observe $Y_i(D_i)$ but not $Y_i(1 - D_i)$; how do we estimate it?

Event Study: Compare $Y_{t+k}(1)$ to $Y_{t-j}(0)$ before and after an event at t .

Matching: Compare $Y_i(1)$ to $Y_j(0)$ for $i \neq j$ but where $X_i = X_j$.

Difference in Difference: Compare $Y_{i,t}(1) - Y_{i,t-1}(0)$ to $Y_{j,t}(0) - Y_{j,t-1}(0)$ for some $j \neq i$.

Synthetic Control: Compare $Y_{i,t}(1) - Y_{i,t-1}(0)$ to $\bar{Y}_{-i,t}(0) - \bar{Y}_{-i,t-1}(0)$ for some average over $j \neq i$ so that $X_{i,t} \approx \bar{X}_{-i,t}$.

Difference in Differences

Motivation: Selection on Observables

Regression and matching on observables (matching: Session 11) have some advantages:

- ▶ Limited assumptions on **functional forms**
- ▶ We could do nearest neighbor matching and use kernels to compute treatment effects

They also have drawbacks:

- ▶ Treated patients are “matched” to control patients based only on **observable characteristics**
 - Ignored **selection on unobservables**.
- ▶ Rely on **cross sectional** variation to construct a control group.

IV estimators resolve some of those issues but

- ▶ Good IV are in short supply!

Often (in this course at least) we have access to **panel data**.

- ▶ What if we could use panel data to control for **unobserved heterogeneity** within a treated individual/group?

Difference in Difference estimators are like the opposite of matching

- ▶ **Strong** assumptions on **functional form**
- ▶ but... allow for **unobservable heterogeneity** in outcomes.

Intro to Difference in Difference

- Consider using OLS to estimate

$$Y_i = \beta_0 + \beta_1 \cdot D_i + \varepsilon_i$$

when $D_i \in \{0, 1\}$ is assigned randomly.

- From the OLS formula and a little algebra, we can show that

$$\hat{b}_1 = \bar{Y}_1 - \bar{Y}_0$$

where $\bar{Y}_1$ is the sample mean of Y_i conditional on $D_i = 1$, and similarly $\bar{Y}_0$ is the sample mean conditional on $D_i = 0$.

- Note: this loops us back to the treatment effects setup, where we were comparing conditional means.

- ▶ **Differences-in-differences** is a **quasi-experimental** design that attempts to deal with selection by focusing on how treatment and control groups change between two periods
- ▶ Consider the model

$$Y_{it} = \beta_0 + \beta_1 \cdot D_{it} + \beta_2' \mathbf{x}_i + \beta_{post} Post_t + \varepsilon_{it},$$

and furthermore suppose that

- Some variables in $\mathbf{X}$ may be unobserved
- D_{it} is not random and may be correlated with $\mathbf{X}_i$
- $Post_t$ is a dummy variable for the post-treatment period $t = 2$
- Each individual i is observed (at least) twice
- For some individuals, D_{it} changes over time (**natural experiment**)
- $\mathbf{X}_i$ does not change over time

$$Y_{it} = \beta_0 + \beta_1 \cdot D_{it} + \beta_2' \mathbf{x}_i + \beta_{post} Post_t + \varepsilon_{it},$$

- Let ΔY_i represent the change in Y_{it} between two time periods:

$$\Delta Y_i = Y_{i2} - Y_{i1},$$

and define ΔD_i and $\Delta \varepsilon_i$ similarly.

- A differenced regression equation:

$$\Delta Y_i = \beta_{post} + \beta_1 \Delta D_i + \Delta \varepsilon_i$$

$$\Delta Y_i = \beta_{post} + \beta_1 \cdot \Delta D_i + \Delta \varepsilon_i$$

- Applying OLS to the differenced regression equation will provide unbiased estimates as long as

$$\mathbb{E} [\Delta \varepsilon_i \mid \Delta D_i] = 0.$$

Exogeneity assumptions of this form are known as the **parallel trends assumption**.

$$\Delta Y_i = \beta_{post} + \beta_1 \cdot \Delta D_i + \Delta \varepsilon_i$$

- ▶ From before, recall that the OLS estimator for a simple experiment amounted to the difference in conditional means of the outcome variable.
- ▶ The same is true here, but now we start with an outcome variable that is differenced over time. Suppose that for some observations, $\Delta D_i = 1$ (treatment group), and for others $D_{i1} = D_{i2} = 0$ (untreated group).

$$\hat{b}_{1,DID} = \overline{\Delta Y}_T - \overline{\Delta Y}_C$$

where $\overline{\Delta Y}_T$ is the conditional mean of ΔY_i for the treatment group, and $\overline{\Delta Y}_C$ is the conditional mean of ΔY_i for the control group.

- ▶ The DID strategy is robust to a form of selection bias (when the selection is related to persistent characteristics). Simple differences across treatment and control groups would not be.
- ▶ DID estimates are also robust to aggregate shocks or time effects. Simple differences over time for the treatment group would not be.
- ▶ DID can be used to study many “natural experiments,” where something changes for one group but not for an otherwise similar group.

A Famous Example: Card and Krueger (AER 1994)

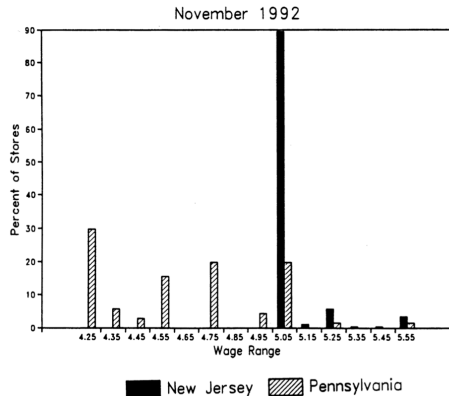
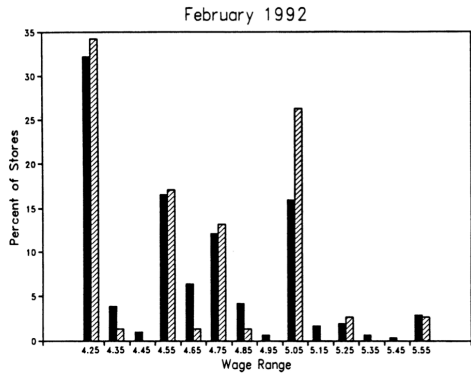
A Famous Example: Card and Krueger (AER 1994)

- ▶ On April 1, 1992 NJ raises its minimum wage from \$4.25 $\rightarrow$ \$5.05 per hour.
- ▶ Question: Econ 101 predicts this will **reduce demand for low wage workers**
 - Focus on fast food restaurants (since they pay min wage)
 - Focus on starting wage (avoid tenure, high turnover)
- ▶ Survey 410 restaurants in NJ (treated group) and eastern PA (control group).
- ▶ Idea: Compare **change** in employment (FTE) in NJ to PA : $\Delta_{DD} = \Delta_{NJ} - \Delta_{PA}$
 - Wave 1: February 15-March 4, 1992
 - Wave 2: November 5 - December 31, 1992

Balance Table: Covariates

Variable	Stores in:		<i>t</i> ^a
	NJ	PA	
1. <i>Distribution of Store Types (percentages):</i>			
a. Burger King	41.1	44.3	−0.5
b. KFC	20.5	15.2	1.2
c. Roy Rogers	24.8	21.5	0.6
d. Wendy's	13.6	19.0	−1.1
e. Company-owned	34.1	35.4	−0.2
2. <i>Means in Wave 1:</i>			
a. FTE employment	20.4 (0.51)	23.3 (1.35)	−2.0
b. Percentage full-time employees	32.8 (1.3)	35.0 (2.7)	−0.7
c. Starting wage	4.61 (0.02)	4.63 (0.04)	−0.4
d. Wage = \$4.25 (percentage)	30.5 (2.5)	32.9 (5.3)	−0.4
e. Price of full meal	3.35 (0.04)	3.04 (0.07)	4.0
f. Hours open (weekday)	14.4 (0.2)	14.5 (0.3)	−0.3
g. Recruiting bonus	23.6 (2.3)	29.1 (5.1)	−1.0
3. <i>Means in Wave 2:</i>			
a. FTE employment	21.0 (0.52)	21.2 (0.94)	−0.2
b. Percentage full-time employees	35.9 (1.4)	30.4 (2.8)	1.8
c. Starting wage	5.08 (0.01)	4.62 (0.04)	10.8
d. Wage = \$4.25 (percentage)	0.0	25.3 (4.9)	—
e. Wage = \$5.05 (percentage)	85.2 (2.0)	1.3 (1.3)	36.1
f. Price of full meal	3.41 (0.04)	3.03 (0.07)	5.0
g. Hours open (weekday)	14.4 (0.2)	14.7 (0.3)	−0.8
h. Recruiting bonus	20.3 (2.3)	23.4 (4.9)	−0.6

Distribution of Wages



Differences in Employment: 2 x 2 Table

TABLE 3—AVERAGE EMPLOYMENT PER STORE BEFORE AND AFTER THE RISE
IN NEW JERSEY MINIMUM WAGE

Variable	Stores by state			Stores in New Jersey ^a			Differences within NJ ^b	
	PA (i)	NJ (ii)	Difference, NJ – PA (iii)	Wage = \$4.25 (iv)	Wage = \$4.26–\$4.99 (v)	Wage ≥ \$5.00 (vi)	Low– high (vii)	Midrange– high (viii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	–2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	–2.69 (1.37)	–2.17 (1.41)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	–0.14 (1.07)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.67 (1.44)	0.75 (1.27)
3. Change in mean FTE employment	–2.16 (1.25)	0.59 (0.54)	2.76 (1.36)	1.32 (0.95)	0.87 (0.84)	–2.04 (1.14)	3.36 (1.48)	2.91 (1.41)
4. Change in mean FTE employment, balanced sample of stores ^c	–2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.21 (0.82)	0.71 (0.69)	–2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0 ^d	–2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	–2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

Notes: Standard errors are shown in parentheses. The sample consists of all stores with available data on employment. FTE (full-time-equivalent) employment counts each part-time worker as half a full-time worker. Employment at six closed stores is set to zero. Employment at four temporarily closed stores is treated as missing.

^aStores in New Jersey were classified by whether starting wage in wave 1 equals \$4.25 per hour ($N = 101$), is between \$4.26 and \$4.99 per hour ($N = 140$), or is \$5.00 per hour or higher ($N = 73$).

^bDifference in employment between low-wage (\$4.25 per hour) and high-wage ($\geq \$5.00$ per hour) stores; and difference in employment between midrange (\$4.26–\$4.99 per hour) and high-wage stores.

^cSubset of stores with available employment data in wave 1 and wave 2.

^dIn this row only, wave-2 employment at four temporarily closed stores is set to 0. Employment changes are based on the subset of stores with available employment data in wave 1 and wave 2.

Outcome Equation

- ▶ Differences lack any covariates (different fast food chains).
- ▶ Also $\Delta_{PA} < 0$ and $\Delta_{NJ} > 0$ (!)
- ▶ Recall i denotes stores, $t \in \{1, 2\}$. Run the following regression:

$$Y_{it} = \beta X_{it} + \alpha \cdot [i \in \text{NJ}] + \gamma \cdot \text{After}_t + \delta \cdot \text{NJ}_i \times \text{After}_t + u_{it}$$

$$Y_{it} = \beta X_{it} + \alpha \cdot [\text{wage gap}_i] + \gamma \cdot \text{After}_t + \delta \cdot \text{wage gap}_i \times \text{After}_t + u_{it}$$

- ▶ α is the $NJ - PA$ difference in period 1
- ▶ γ is the change from period 1 to 2 in PA (control)
- ▶ δ is the parameter of interest, the **difference in difference**
- ▶ $\text{wage gap}_i = \max\{0, 5.05 - w_{i1}\}/w_{i1}$ for NJ stores, 0 for PA.
(By what proportion must the $t = 1$ wage rise to reach the new minimum?)

TABLE 4—REDUCED-FORM MODELS FOR CHANGE IN EMPLOYMENT

Independent variable	Model				
	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	2.33 (1.19)	2.30 (1.20)	—	—	—
2. Initial wage gap ^a	—	—	15.65 (6.08)	14.92 (6.21)	11.91 (7.39)
3. Controls for chain and ownership ^b	no	yes	no	yes	yes
4. Controls for region ^c	no	no	no	no	yes
5. Standard error of regression	8.79	8.78	8.76	8.76	8.75
6. Probability value for controls ^d	—	0.34	—	0.44	0.40

Notes: Standard errors are given in parentheses. The sample consists of 357 stores with available data on employment and starting wages in waves 1 and 2. The dependent variable in all models is change in FTE employment. The mean and standard deviation of the dependent variable are -0.237 and 8.825 , respectively. All models include an unrestricted constant (not reported).

^aProportional increase in starting wage necessary to raise starting wage to new minimum rate. For stores in Pennsylvania the wage gap is 0.

^bThree dummy variables for chain type and whether or not the store is company-owned are included.

^cDummy variables for two regions of New Jersey and two regions of eastern Pennsylvania are included.

^dProbability value of joint F test for exclusion of all control variables.

Inference with Few Treated Clusters

- ▶ Card and Krueger's SEs treat 410 stores as independent, but the policy varies by *state*: NJ vs. PA.
- ▶ Stores in a state share shocks (local demand, weather). Cluster-robust SEs need many clusters, and many *treated* clusters; with $G = 2$ you cannot cluster by state at all.
- ▶ The wild cluster bootstrap (Cameron, Gelbach and Miller 2008) fixes *few clusters*, not *few treated clusters*: with one or two treated clusters it still over-rejects (MacKinnon and Webb 2018).
- ▶ With one treated group and *many* control groups:
 - Conley and Taber (2011): control groups' residuals estimate the distribution of the treated group's error.
 - Randomization inference: assign the policy to each control group in turn (the placebo logic of Session 11).
 - Aggregate to group $\times$ period and use $G - 1$ degrees of freedom (Donald and Lang 2007).
- ▶ With one control group, as in Card and Krueger, none of these can deliver policy-level inference. Count your treated (and control) clusters before you report a standard error.

Parallel trends vs pre-trends

- ▶ When the data is available, researchers typically check **pre-trends** as a way of motivating the parallel trends assumption.
 - That is, look at how the control and treatment groups evolve in several periods preceding the time of treatment to see if they were changing in similar ways over time.
- ▶ This is a worthwhile check, and we should certainly worry about the plausibility of the parallel trends assumption if the pre-trends are not parallel.
- ▶ However, the parallel trends assumption is about what *would have happened* immediately at the time of treatment in the absence of treatment, not about what happened before treatment. Thus, establishing parallel pre-trends **does not** establish the parallel trends assumption.
 - Just as we can't test exogeneity within the OLS framework, we can't test the parallel trends assumption within the DiD framework.

Parallel trends vs pre-trends: example

Consider the NJ minimum wage study. Here's a concern that would not be assuaged by seeing parallel pre-trends:

- ▶ If the minimum wage changed because a new administration was elected in NJ and they implemented a portfolio of new policies, then the minimum wage change might have been just one of several ways in which NJ and PA were diverging at the time.
- ▶ Some of the other policy changes might have led to greater labor force participation and/or incentivized businesses to hire more.
- ▶ Because of these other changes, we would expect NJ's restaurant employment to move differently from PA's in 1992 even if the minimum wage change had not happened.
- ▶ This could all be true even if NJ and PA were evolving similarly before 1992. The point is that the treatment was one thing that diverged in 1992, and if one thing was diverging at that point, perhaps other things were also diverging.

- ▶ Recall that before we argued that experimental effects can be estimated with regressions using additional covariates (controls) even though they don't need to be.
- ▶ Similarly, they can be estimated using differences-in-differences rather than simple differences. They don't need to be: with randomized treatment, the treatment and control groups should have similar baselines.

- ▶ The study has been controversial, and it has had a big impact on policy discussions.
- ▶ Card and Krueger update their conclusion in a 2000 follow-up:
The increase in New Jersey's minimum wage probably had no effect on total employment in New Jersey's fast-food industry, and possibly had a small positive effect.
- ▶ There have been *many* criticisms of the paper, including studies showing that the result is not robust to the sample of stores and data source used, and concerns about the external validity of the focus on only fast food stores.
- ▶ Also see studies based on Seattle's 2015 minimum wage change.

A More General Method

Difference in Difference: Setup

Observed outcome in period 1 = $Y_{i1}(0)$

Observed outcome in period 2 = $\left\{ \begin{array}{ll} Y_{i2}(1) & \text{if } D_{i2} = 1 \\ Y_{i2}(0) & \text{if } D_{i2} = 0 \end{array} \right\}$

	Treatment	Control
Before	$Y_{i1}(0)$	$Y_{i1}(0)$
After	$Y_{i2}(1)$	$Y_{i2}(0)$

We can write the outcome as:

$$Y_{it} = D_{it} \cdot Y_{it}(1) + (1 - D_{it}) \cdot Y_{it}(0) = D_{it} \cdot (Y_{it}(1) - Y_{it}(0)) + Y_{it}(0)$$

Difference in Difference: Decomposition

Consider the first difference $\Delta Y_{it} = Y_{i2} - Y_{i1}$:

$$\Delta Y_{it} = D_{i2} \cdot (Y_{i2}(1) - Y_{i2}(0)) + Y_{i2}(0) - Y_{i1}(0)$$

For treated group (first difference):

$$\mathbb{E}[\Delta Y_{it} \mid D_{i2} = 1] = \overbrace{\mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid D_{i2} = 1]}^{ATT} + \overbrace{\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_{i2} = 1]}^{\gamma(1)}$$

For control group (second difference):

$$\mathbb{E}[\Delta Y_{it} \mid D_{i2} = 0] = \overbrace{\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_{i2} = 0]}^{\gamma(0)}$$

The DiD (difference in difference) estimator

$$\Delta_{DD} = \mathbb{E}[\Delta Y_{it} \mid D_{i2} = 1] - \mathbb{E}[\Delta Y_{it} \mid D_{i2} = 0]$$

Difference in Difference: Parallel Trends

- ▶ If $\gamma(1) = \gamma(0)$ then the trend terms cancel and we are left with $\Delta_{DD} = \text{ATT}$.
- ▶ This is the **parallel trends assumption**

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_{i2} = 0] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_{i2} = 1]$$

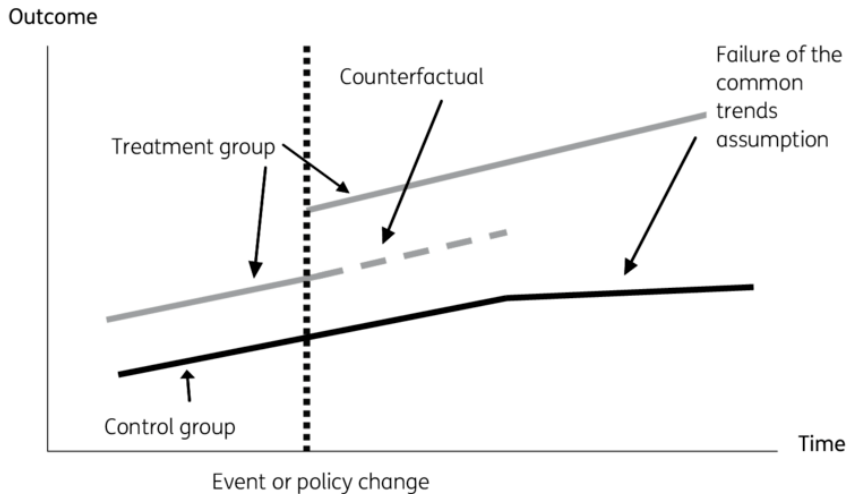
- ▶ Absent the treatment effect, both treatment and control would evolve identically over time.
- ▶ But, treatment and control groups can start from very different places...

$$\mathbb{E}[Y_{it}(0) \mid D_{i2} = 1] \neq \mathbb{E}[Y_{it}(0) \mid D_{i2} = 0], t = 1, 2$$

- ▶ And have selection on treatment effects...

$$\mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid D_{i2} = 1] \neq \mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid D_{i2} = 0]$$

Parallel Trends



Difference in Differences: Limitations

1. Functional form restrictions

- **Parallel trends** assumes that, absent treatment, we add $\gamma_2 - \gamma_1$ to each unit
- Because this is **additive** it is not invariant to transformations $f(Y_{it})$ (e.g., taking logs)

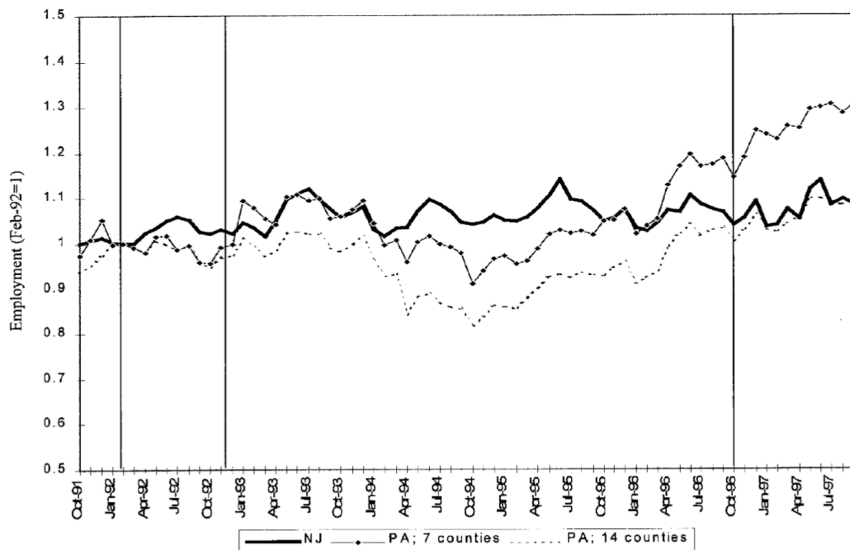
2. Parallel Trend Assumption is **not testable**

- Best we can hope is that it looks similar in the pre-period

3. Compositional Effects: the treatment may affect who is in each group

- Restaurants could close in NJ and open nearby in PA to avoid minimum wage.
- A good job training program may lead to migration, etc.
- One approach: redefine the population so that it doesn't endogenously respond to treatment
 - Recover something, but probably not ATT anymore...

Checking Pre-Trend: Card and Krueger (2000)



Difference in Differences: Regression Form

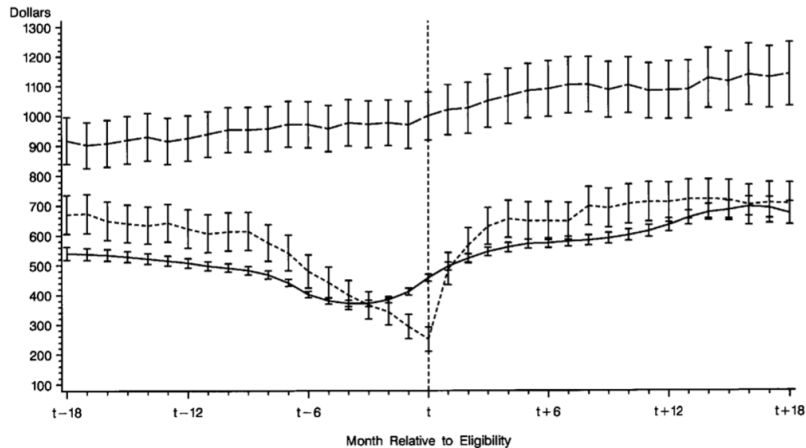
Just like in Card and Krueger, we can write as a regression equation:

$$Y_{it} = \alpha_i + \gamma_t + \beta X_{it} + \delta_i D_{it} + u_{it}$$

- ▶ Suppose we wish to evaluate a training program for those with low earnings. Let the threshold for eligibility be B .
- ▶ We have a panel of individuals and those with low earnings qualify for training, forming the treatment group.
- ▶ Those with higher earnings form the control group.
- ▶ Now the low earning group is low for two reasons
 1. They have low permanent earnings (α_i is low) - this is accounted for by diff in diffs.
 2. They have a negative transitory shock (u_{i1} is low) - this is not accounted for by diff in diffs.

The “Ashenfelter Dip” (Heckman and Smith 1999)

FIGURE 1A
MEAN SELF-REPORTED MONTHLY EARNINGS
SIPP Eligibles and JTPA Controls and ENPs
Male Adults



- ▶ #2 above violates the assumption $\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0)]$.
- ▶ To see why note that those participating in the program are such that $Y_{i1}(0) < B$. Assume for simplicity that the shocks u are *iid*. Hence $u_{i1} < B - \alpha_i - \gamma_1$. This implies:

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D = 1] = \gamma_2 - \gamma_1 - \mathbb{E}[u_{i1} \mid u_{i1} < B - \alpha_i - \gamma_1]$$

For the control group:

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D = 0] = \gamma_2 - \gamma_1 - \mathbb{E}[u_{i1} \mid u_{i1} > B - \alpha_i - \gamma_1]$$

So that:

$$\begin{aligned} \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D = 1] - \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D = 0] = \\ \mathbb{E}[u_{i1} \mid u_{i1} > B - \alpha_i - \gamma_1] - \mathbb{E}[u_{i1} \mid u_{i1} < B - \alpha_i - \gamma_1] > 0 \end{aligned}$$

- ▶ This is effectively regression to the mean: those unlucky enough to have a bad shock recover and show greater growth relative to those with a good shock. The nature of the bias depends on the stochastic properties of the shocks and how individuals select into training.

Conditional Parallel Trends

- ▶ The assumption on growth of the non-treatment outcome being independent of assignment to treatment may be violated, but it may still be true conditional on X .
- ▶ Consider the assumption

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid X, D] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid X]$$

- ▶ This is just the matching assumption on a redefined variable, namely the growth in the outcomes. In its simplest form the approach is implemented by running the regression

$$Y_{it} = \alpha_i + \gamma_t + \delta_i D_{it} + \beta'_t X_i + u_{it}$$

which allows for differential trends in the non-treatment growth depending on X_i . More generally one can implement propensity score matching on the growth of the outcome variable when panel data is available.

Variants

The triple difference is also a thing:

- ▶ Suppose that we have: before/after, treated-state/untreated-state, treated-group (men)/untreated-group (women).
- ▶ We can compute two D-i-D here: $\Delta_{DDD} = \Delta_{DD}^{\text{men}} - \Delta_{DD}^{\text{women}}$, where each is the treated-vs-untreated-state DiD within that group.
- ▶ Literally the difference of two difference-in-differences estimators.
- ▶ As a regression: interpret the triple-interaction term (make sure to control for ALL double interactions).

- Suppose individuals have time-varying observable covariates:

$$Y_{it} = \beta_0 + \beta_1 \cdot D_{it} + \beta_2' \mathbf{x}_{it} + \varepsilon_{it},$$

- We can still estimate β_1 with a differenced linear regression,

$$\Delta Y_i = \beta_1 \cdot \Delta D_i + \beta_2' \Delta \mathbf{x}_i + \Delta \varepsilon_i$$

noting that the exogeneity assumption becomes

$$\mathbb{E} \left[\Delta \varepsilon_i \middle| \begin{pmatrix} \Delta D_i \\ \Delta \mathbf{x}_i \end{pmatrix} \right] = 0$$

- The estimate of β_1 here does not correspond to a simple difference in differences, but people still describe this sort of estimation strategy as DID. Given Frisch-Waugh, it's still DID after controlling for $\mathbf{X}$.

- ▶ A **staggered diff-in-diff** involves different groups being treated at different points in time. Typically, these models are estimated with a linear regression with two-way fixed effects. (More on this in Lecture 9b.)
- ▶ See [Callaway and Sant'Anna \(2021\)](#), [de Chaisemartin and D'Haultfoeuille \(2023\)](#) on why this is not a valid strategy in the presence of heterogeneous treatment effects and for alternative estimators. (More on this in Lecture 9b.)